

Load the packages

```
using DataFramesMeta
using CSV
using ReadStatTables
using Statistics
using CairoMakie
using AlgebraOfGraphics
using PairPlots
using ColorSchemes
using StyledStrings
using SummaryTables
using PairPlots
include(joinpath(@__DIR__, "table-one-mod.jl"))
include(joinpath(@__DIR__, "pairplot-with-regression.jl"))
include(joinpath(@__DIR__, "boxplot-with-stats.jl"))
include(joinpath(@__DIR__, "coerce_string_columns.jl"))
include(joinpath(@__DIR__, "eta-corr.jl"))
include(joinpath(@__DIR__, "gof.jl"))
include(joinpath(@__DIR__, "indiv-fits.jl"))
include(joinpath(@__DIR__, "model-comparison-table.jl"))
# include(joinpath(@__DIR__, "ParameterTables.jl"))
include(joinpath(@__DIR__, "plot-fun.jl"))
include(joinpath(@__DIR__, "session-info.jl"))
```

```
Main.Notebook.sessioninfo
```

Initialize the dataset and result directories

```
# Set relative paths to directories (start from the project folder)
proj_dir = string(chomp(read(`git rev-parse --show-toplevel`, String))) # Get
the root git folder
pumasai_repo = basename(realpath(proj_dir))
data_dir = joinpath(proj_dir, "datasets")
results_dir = joinpath(proj_dir, "results")
sdata_dir = joinpath(data_dir, "source")
ddata_dir = joinpath(data_dir, "derived")

# Specify the Quarto file specific directories
qmd_dir, qmd_name = splitdir(@__FILE__)
qmd_prefix, _ = splitext(qmd_name)
task_results_dir = joinpath(results_dir, basename(realpath(qmd_dir)))
qmd_results_dir = joinpath(task_results_dir, qmd_prefix)
mdl_dir = joinpath(qmd_dir, "pumas")
julia_fns_dir = joinpath(qmd_dir, "julia-functions")

# Specify the qmd relative path to program and default source info
qmd_relative_path = lstrip(split(@__FILE__, pumasai_repo)[end], '/')
```

```
# Initialize results directory
# include(joinpath(utils_dir, "init-dir.jl"))
# init_dir(joinpath(qmd_results_dir))
```

```
"\\programs\\Plots\\plot-adppk.qmd"
```

Specify text strings to be used as program and source references in footnotes to tables and figures created by this Quarto program.

```
# Text specifying relative paths to program and default source info
# Program path
prog_ref = string("Program: ", qmd_relative_path)

# Relative path to Source path: QMD HTML
qmd_html = string(qmd_prefix, ".html")
qmd_html_path = joinpath(
    dirname(qmd_relative_path),
    qmd_html
)
qmd_html_ref = string("Source: ", qmd_html_path)

# Specify relative path to the QMD results directory
# This should be used to construct the Source reference for TLFs
qmd_results_path = replace(qmd_relative_path, "programs" => "results")
```

```
"\\results\\Plots\\plot-adppk.qmd"
```

Setting Plots theme

```
my_theme = Theme(
  Axis=(
    yminorticks=IntervalsBetween(2),
    yminorgridvisible=true,
    xminorticks=IntervalsBetween(2),
    xminorgridvisible=true,
    xlabelfont=:bold,
    ylabelfont=:bold,
    leftspinecolor=:gray60,
    rightspinecolor=:gray60,
    topspinecolor=:gray60,
    bottomspinecolor=:gray60,
  )
)
```

```

my_log_theme = Theme(
  Axis=(
    yminorgridvisible=true,
    xminorticks=IntervalsBetween(2),
    xminorgridvisible=true,
    xlabelfont=:bold,
    ylabelfont=:bold,
    leftspinecolor=:gray60,
    rightspinecolor=:gray60,
    topspinecolor=:gray60,
    bottomspinecolor=:gray60,
    yscale=log10,
    yminorticks=IntervalsBetween(10),
    ytickformat=x -> string.(round.(x, digits=1)),
  )
)

set_theme!(my_theme)

```

Load the data

```

path = joinpath(@__DIR__ , "..", "..", "datasets", "derived")
adppk = CSV.read(
  joinpath(path, "adppk.csv"),
  DataFrame
)
adppk = filter(:COHORTC => !=("Placebo"), adppk)
first(adppk, 5)

```

```

5×63 DataFrame
 Row | STUDYID      USUBJID      ADY   EVID   NFRLT   HOURS_BETWEEN_DOSES
...  |          |          |   |   |   |
...  | String15   String15   Int64 Int64  Float64 Int64?
...  |-----|-----|
  1 | CDISCPIL0T01 01-710-1002    0    0    0.0      missing
...
  2 | CDISCPIL0T01 01-710-1002    1    1    0.0         24
  3 | CDISCPIL0T01 01-710-1002    1    0    0.08      missing
  4 | CDISCPIL0T01 01-710-1002    1    0    0.5      missing
  5 | CDISCPIL0T01 01-710-1002    1    0    1.0      missing
...
                                         57 columns

omitted

```

Dataset to be inspected interactively, to ensure consistency with the expected structure and content.

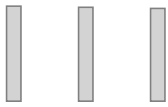
```
println("No. of rows in the DataFrame: ", nrow(adppk))
println("No. of unique study participants: ", length(unique(adppk.USUBJID)))
println("Total No. of Observations: ", count(==(0), adppk.EVID))
```

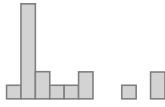
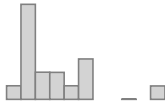
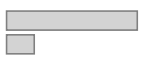
```
No. of rows in the DataFrame: 2850
No. of unique study participants: 168
Total No. of Observations: 2352
```

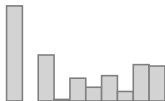
Dataset overview

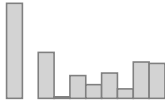
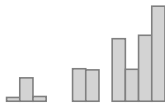
```
tbl = overview_table(adppk)
io = IOBuffer()
show(io, MIME"text/html"(), tbl)
html_output = String(take!(io))
write(joinpath(@__DIR__ , "..", "..", "datasets", "derived",
"poppk_overview.html"), html_output)
tbl
```

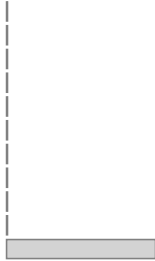
No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
1	STUDYID [String15]	1. CDISCPLOT02850 (100%)			2850 (100%)	0 (0%)
		1. 01-701-1429				
		2. 01-705-1031	17 (0.6%)			
		3. 01-713-1209	17 (0.6%)			
		4. 01-701-1115	17 (0.6%)			
2	USUBJID [String15]	5. 01-703-1086	17 (0.6%)		2850 (100%)	0 (0%)
		6. 01-704-1065	17 (0.6%)			
		7. 01-706-1049	2680 (94%)			
		8. 01-701-1341				
		9.				

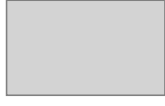
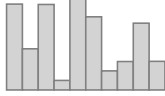
No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
3	ADY [Int64]	01-701-1442 10. 01-701-1033 [158 others] Mean (sd): 1.35 (0.761) min ≤ med ≤ max: 0 ≤ 1 ≤ 3 IQR (CV): 1 (0.563)	4 distinct values		2850 (100%)	0 (0%)
4	EVID [Int64]	Mean (sd): 0.175 (0.38) min ≤ med ≤ max: 0 ≤ 0 ≤ 1 IQR (CV): 0 (2.17)	2 distinct values		2850 (100%)	0 (0%)
5	NFRLT [Float64]	Mean (sd): 13.5 (16.1) min ≤ med ≤ max: 0 ≤ 6 ≤ 48 IQR (CV): 23 (1.19)	14 distinct values		2850 (100%)	0 (0%)
6	HOURS_BETWEEN_DOSES [Int64?]	Mean (sd): 24 (0) min ≤ med ≤ max: 24 ≤ 24 ≤ 24 IQR (CV): 0 (0)	1 distinct values		498 (17.5%)	2352 (82.5%)
7	DOSE_INDEX [Int64?]	Mean (sd): 1.99 (0.817) min ≤ med ≤ max: 1 ≤ 2 ≤ 3	3 distinct values		498 (17.5%)	2352 (82.5%)

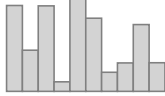
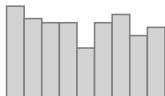
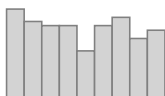
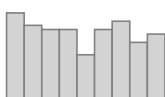
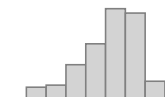
No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
8	AFRLT [Float64?]	IQR (CV): 2 (0.41) Mean (sd): 13.5 (16.1) min ≤ med ≤ max: -0.5 ≤ 6 ≤ 48 IQR (CV): 23 (1.2) Mean (sd): 10.7 (12.7)	15 distinct values		2850 (100%)	0 (0%)
9	APRLT [Float64?]	min ≤ med ≤ max: -0.5 ≤ 6 ≤ 48 IQR (CV): 15 (1.18) Mean (sd): 54 (0)	15 distinct values		2850 (100%)	0 (0%)
10	DOSEA [Float64?]	min ≤ med ≤ max: 54 ≤ 54 ≤ 54 IQR (CV): 0 (0) Mean (sd): 67.5 (13.5)	1 distinct values		2682 (94.1%)	168 (5.9%)
11	DOSEP [Int64]	min ≤ med ≤ max: 54 ≤ 54 ≤ 81 IQR (CV): 27 (0.2)	2 distinct values		2850 (100%)	0 (0%)
12	PARAMCD [String7]	1. XAN 2. DOSE	2352 (82.5%) 498 (17.5%)		2850 (100%)	0 (0%)
13	ALLOQ [Float64?]	Mean (sd): 0.01 (1.74e-18) min ≤ med ≤ max:	1 distinct values		2352 (82.5%)	498 (17.5%)

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
14	CMT [Int64]	$0.01 \leq 0.01 \leq 0.01$ IQR (CV): 0 (1.74e-16) Mean (sd): 1.83 (0.38) min $\leq$ med $\leq$ max: $1 \leq 2 \leq 2$ IQR (CV): 0 (0.208)	2 distinct values		2850 (100%)	0 (0%)
15	BLQFL [String1]	1. N 2. Y Mean (sd): 0.177 (0.382) min $\leq$ med $\leq$ max: $0 \leq 0 \leq 1$ IQR (CV): 0 (2.16)	2346 (82.3%) 504 (17.7%)		2850 (100%)	0 (0%)
16	BLQFN [Int64]	Mean (sd): 0.177 (0.382) min $\leq$ med $\leq$ max: $0 \leq 0 \leq 1$ IQR (CV): 0 (2.16)	2 distinct values		2850 (100%)	0 (0%)
17	AMT [Float64?]	Mean (sd): 54 (0) min $\leq$ med $\leq$ max: $54 \leq 54 \leq 54$ IQR (CV): 0 (0)	1 distinct values		498 (17.5%)	2352 (82.5%)
18	DV [Float64?]	Mean (sd): 0.858 (0.694) min $\leq$ med $\leq$ max: $0 \leq 0.736 \leq 1.94$ IQR (CV): 1.41 (0.81)	1849 distinct values		2016 (70.7%)	834 (29.3%)

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
19	AVAL [Float64?]	Mean (sd): 0.858 (0.694) min ≤ med ≤ max: 0 ≤ 0.736 ≤ 1.94 IQR (CV): 1.41 (0.81)	1849 distinct values		2016 (70.7%)	834 (29.3%)
20	DVL [Float64?]	Mean (sd): -0.669 (1.47) min ≤ med ≤ max: -4.6 ≤ -0.0764 ≤ 0.662 IQR (CV): 2.25 (-2.19)	1848 distinct values		1848 (64.8%)	1002 (35.2%)
21	MDV [Int64]	Mean (sd): 0.293 (0.455) min ≤ med ≤ max: 0 ≤ 0 ≤ 1 IQR (CV): 1 (1.56)	2 distinct values		2850 (100%)	0 (0%)
22	AVALU [String7?]	1. ug/ml	2352 (100%)		2352 (82.5%)	498 (17.5%)
23	UDTC [DateTime]	1. Dates.DateTime7("2013-10-26T00:00:00") 2. Dates.DateTime6("2013-01-07T00:00:00") 3. Dates.DateTime6("2012-12-02T00:00:00") 4. Dates.DateTime6("2013-09-17T00:00:00") 5. Dates.DateTime6("2013-02-17T00:00:00") 6. Dates.DateTime6("2013-10-27T00:00:00") 7. Dates.DateTime6("2013-05-15T00:00:00") 8. Dates.DateTime6("2012-09-09T00:00:00") 9. Dates.DateTime6("2013-01-06T00:00:00") 10. Dates.DateTime6("2013-12-30T00:00:00") [2035 others]	2789 (97.9%)		2850 (100%)	0 (0%)

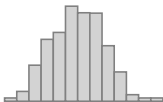
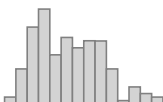
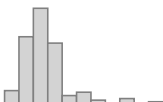
No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
24	II [Int64]	Mean (sd): 0.175 (0.38) min ≤ med ≤ max: 0 ≤ 0 ≤ 1 IQR (CV): 0 (2.17)	2 distinct values		2850 (100%)	0 (0%)
25	SS [Int64]	Mean (sd): 0.175 (0.38) min ≤ med ≤ max: 0 ≤ 0 ≤ 1 IQR (CV): 0 (2.17)	2 distinct values		2850 (100%)	0 (0%)
26	ASEQ [String]	1. Base.EachStringIndex{String} ("01-708-1272") 2. Base.EachStringIndex{String} ("01-716-1311") 3. Base.EachStringIndex{String} ("01-708-1372") 17 (0.6%) 4. Base.EachStringIndex{String} ("01-703-1335") 17 (0.6%) 5. Base.EachStringIndex{String} ("01-713-1043") 17 (0.6%) 6. Base.EachStringIndex{String} ("01-718-1427") 17 (0.6%) 7. Base.EachStringIndex{String} ("01-716-1103") 17 (0.6%) 8. Base.EachStringIndex{String} ("01-710-1300") 2680 (94%) 9. Base.EachStringIndex{String} ("01-701-1341") 10. Base.EachStringIndex{String} ("01-708-1348") [158 others]			2850 (100%)	0 (0%)
27	PARAM [String]	1. Pharmacokinetic concentration of	2352 (82.5%) 498 (17.5%)		2850 (100%)	0 (0%)

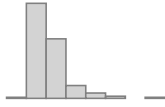
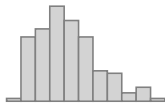
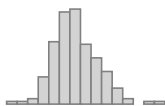
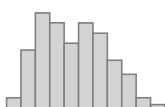
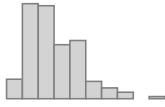
No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
28	PARAMN [Int64]	Xanomeline 2. Xanome- line Patch Dose Mean (sd): 1.17 (0.38) min ≤ med ≤ max: 1 ≤ 1 ≤ 2 IQR (CV): 0 (0.323)	2 distinct val- ues		2850 (100%)	0 (0%)
29	PROJID [String15]	1. XANOME- LINE Mean (sd): 1 (0) min ≤ med ≤ max: 1 ≤ 1 ≤ 1 IQR (CV): 0 (0)	2850 (100%)		2850 (100%)	0 (0%)
30	PROJIDN [Int64]	Mean (sd): 1 (0) min ≤ med ≤ max: 1 ≤ 1 ≤ 1 IQR (CV): 0 (0)	1 distinct val- ues		2850 (100%)	0 (0%)
31	STUDYIDN [Int64]	Mean (sd): 1 (0) min ≤ med ≤ max: 1 ≤ 1 ≤ 1 IQR (CV): 0 (0)	1 distinct val- ues		2850 (100%)	0 (0%)
32	SITEID [Int64]	Mean (sd): 708 (5.45) min ≤ med ≤ max: 701 ≤ 708 ≤ 718 IQR (CV): 9 (0.0077)	17 distinct values		2850 (100%)	0 (0%)

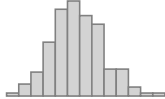
No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
33	SITEIDN [Int64]	Mean (sd): 708 (5.45) min ≤ med ≤ max: 701 ≤ 708 ≤ 718 IQR (CV): 9 (0.0077)	17 distinct values		2850 (100%)	0 (0%)
34	USUBJIDN [Int64]	Mean (sd): 1217 (133) min ≤ med ≤ max: 1002 ≤ 1213 ≤ 1448 IQR (CV): 229 (0.109)	168 distinct values		2850 (100%)	0 (0%)
35	SUBJID [Int64]	Mean (sd): 1217 (133) min ≤ med ≤ max: 1002 ≤ 1213 ≤ 1448 IQR (CV): 229 (0.109)	168 distinct values		2850 (100%)	0 (0%)
36	SUBJIDN [Int64]	Mean (sd): 1217 (133) min ≤ med ≤ max: 1002 ≤ 1213 ≤ 1448 IQR (CV): 229 (0.109)	168 distinct values		2850 (100%)	0 (0%)
37	AGE [Float64]	Mean (sd): 75 (8.07) min ≤ med ≤ max: 51 ≤ 77 ≤ 88	31 distinct values		2850 (100%)	0 (0%)

No	Variable	Stats / Val- ues	Freqs (% of Valid)	Graph	Valid	Missing	
		IQR (CV): 10 (0.108)					
38	SEX [String1]	1. F 2. M	1528 (53.6%) 1322 (46.4%)		2850 (100%)	0 (0%)	
		Mean (sd): 1.54 (0.499)					
39	SEXN [Int64]	min ≤ med ≤ max: 1 ≤ 2 ≤ 2	2 distinct val- ues			2850 (100%)	0 (0%)
		IQR (CV): 1 (0.325)					
		Mean (sd): 1.5 (0.5)					
40	COHORT [Int64]	min ≤ med ≤ max: 1 ≤ 1 ≤ 2	2 distinct val- ues			2850 (100%)	0 (0%)
		IQR (CV): 1 (0.334)					
41	COHORTC [String31]	1. Xanome- line Low Dose 2. Xanome- line High Dose	1427 (50.1%) 1423 (49.9%)		2850 (100%)	0 (0%)	
42	ROUTE [String15]	1. TRANS- DERMAL	2850 (100%)		2850 (100%)	0 (0%)	
		Mean (sd): 3 (0)					
43	ROUTEN [Int64]	min ≤ med ≤ max: 3 ≤ 3 ≤ 3	1 distinct val- ues			2850 (100%)	0 (0%)
		IQR (CV): 0 (0)					
44	RACE [String]	1. WHITE 2. BLACK OR AFRICAN AMERICAN	2578 (90.5%) 255 (8.9%) 17 (0.6%)		2850 (100%)	0 (0%)	

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
45	RACEN [Int64]	3. AMERICAN INDIAN OR ALASKA NATIVE Mean (sd): 4.8 (0.642) min ≤ med ≤ max: 1 ≤ 5 ≤ 5 IQR (CV): 0 (0.134)	3 distinct values		2850 (100%)	0 (0%)
46	ETHNIC [String31]	1. NOT HISPANIC OR LATINO 2. HISPANIC OR LATINO Mean (sd): 1.95 (0.225) min ≤ med ≤ max: 1 ≤ 2 ≤ 2 IQR (CV): 0 (0.116)	2697 (94.6%) 153 (5.4%)		2850 (100%)	0 (0%)
47	ETHNICN [Int64]	1. PATCH Mean (sd): 3 (0) min ≤ med ≤ max: 3 ≤ 3 ≤ 3 IQR (CV): 0 (0)	2 distinct values		2850 (100%)	0 (0%)
48	FORM [String7]	1. PATCH Mean (sd): 3 (0) min ≤ med ≤ max: 3 ≤ 3 ≤ 3 IQR (CV): 0 (0)	2850 (100%)		2850 (100%)	0 (0%)
49	FORMN [Int64]	1. PATCH Mean (sd): 3 (0) min ≤ med ≤ max: 3 ≤ 3 ≤ 3 IQR (CV): 0 (0)	1 distinct values		2850 (100%)	0 (0%)
50	COUNTRY [String3]	1. USA Mean (sd): 3 (0) min ≤ med ≤ max: 3 ≤ 3 ≤ 3 IQR (CV): 0 (0)	2850 (100%)		2850 (100%)	0 (0%)
51	country_name [String31]	1. United States of America Mean (sd): 3 (0) min ≤ med ≤ max: 3 ≤ 3 ≤ 3 IQR (CV): 0 (0)	2850 (100%)		2850 (100%)	0 (0%)

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
52	country_number [Float64]	Mean (sd): 235 (0) min ≤ med ≤ max: 235 ≤ 235 ≤ 235 IQR (CV): 0 (0)	1 distinct values		2850 (100%)	0 (0%)
53	HTBL [Float64]	Mean (sd): 165 (10.3) min ≤ med ≤ max: 136 ≤ 165 ≤ 196 IQR (CV): 14 (0.0625)	47 distinct values		2850 (100%)	0 (0%)
54	WTBL [Float64?]	Mean (sd): 68.6 (14.4) min ≤ med ≤ max: 41.7 ≤ 68 ≤ 108 IQR (CV): 22.2 (0.209)	95 distinct values		2833 (99.4%)	17 (0.6%)
55	ALTBL [Float64?]	Mean (sd): 18.6 (9.38) min ≤ med ≤ max: 5 ≤ 17 ≤ 70 IQR (CV): 6 (0.505)	32 distinct values		2816 (98.8%)	34 (1.2%)
56	ASTBL [Float64?]	Mean (sd): 23.3 (7.42) min ≤ med ≤ max: 13 ≤ 21 ≤ 60 IQR (CV): 7 (0.319)	28 distinct values		2816 (98.8%)	34 (1.2%)

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
57	TBILBL [Float64?]	Mean (sd): 10.2 (4.78) min ≤ med ≤ max: 3.42 ≤ 8.55 ≤ 39.3 IQR (CV): 5.13 (0.467)	15 distinct values		2816 (98.8%)	34 (1.2%)
58	CREATBL [Float64?]	Mean (sd): 104 (19.6) min ≤ med ≤ max: 61.9 ≤ 97.2 ≤ 168 IQR (CV): 26.5 (0.189)	13 distinct values		2816 (98.8%)	34 (1.2%)
59	BMIBL [Float64?]	Mean (sd): 25.2 (4.19) min ≤ med ≤ max: 13.7 ≤ 24.8 ≤ 40.2 IQR (CV): 5.23 (0.166)	160 distinct values		2833 (99.4%)	17 (0.6%)
60	BSABL [Float64?]	Mean (sd): 1.76 (0.222) min ≤ med ≤ max: 1.38 ≤ 1.77 ≤ 2.31 IQR (CV): 0.38 (0.126)	160 distinct values		2833 (99.4%)	17 (0.6%)
61	CRCLBL [Float64?]	Mean (sd): 50.5 (15.9) min ≤ med ≤ max: 24.2 ≤ 47.1 ≤ 116	165 distinct values		2799 (98.2%)	51 (1.8%)

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
62	EGFRBL [Float64?]	IQR (CV): 22.5 (0.315)				
		Mean (sd): 54.2 (10.6)				
		min ≤ med ≤				
		max: 135 distinct values			2816 (98.8%)	34 (1.2%)
63	RECSEQ [String]	28.4 ≤ 53.8 ≤ 87.2				
		IQR (CV): 13.8 (0.196)				
		1. Base.EachStringIndex{String} ("01-708-1272")				
		2. Base.EachStringIndex{String} ("01-716-1311")				
		3. Base.EachStringIndex{String} ("01-708-1372") 17 (0.6%)				
		4. Base.EachStringIndex{String} ("01-703-1335") 17 (0.6%)				
		5. Base.EachStringIndex{String} ("01-713-1043") 17 (0.6%)				
		6. Base.EachStringIndex{String} ("01-718-1427") 17 (0.6%)				
		7. Base.EachStringIndex{String} ("01-716-1103") 17 (0.6%)				
		8. Base.EachStringIndex{String} ("01-710-1300") 2680 (94%)			2850 (100%)	0 (0%)
		9. Base.EachStringIndex{String} ("01-701-1341")				
		10. Base.EachStringIndex{String} ("01-708-1348")				
		[158 others]				

Dimensions: 2850 x 63

Duplicate rows: 0

Tabular Summary of Subject Characteristics

```
# Baseline characteristics
# Note: Missing row appears only when there are missing values
adata_subj = unique(adppk, :USUBJID) # Subset to one row per subject (USUBJID)
```

```

table_one_mod( # bespoke function table_one_mod - renders Total column in the
last. Use table_one to get it in the front.
  adata_subj,
  [
    :SEX => "Sex",
    :AGE => "Age [yr]",
    :WTBL => "Baseline Weight [kg]",
    :BMIBL => "Baseline BMI [kg/m²]",
    :EGFRBL => "Baseline eGFR (CKD-2021) [mL/min/1.73 m²]"
  ],
  groupby=:STUDYID => "Study",
  show_n=true,
  footnotes=[
    "Abbreviations: BMI = body mass index, CKD = chronic kidney disease,
    eGFR = estimated glomerular filtration rate, max = maximum, min =
    minimum, SD = Standard Deviation",
    "Note: Missing row appears only when there are missing values"
  ]
)

```

	Study	
	CDISCPLOT01 (n=168)	Total (n=168)
Sex		
F	90 (53.6%)	90 (53.6%)
M	78 (46.4%)	78 (46.4%)
Age [yr]		
Mean (SD)	75 (8.09)	75 (8.09)
Median [Min, Max]	77 [51, 88]	77 [51, 88]
Baseline Weight [kg]		
Mean (SD)	68.6 (14.4)	68.6 (14.4)
Median [Min, Max]	68 [41.7, 108]	68 [41.7, 108]
Missing	1 (0.595%)	1 (0.595%)
Baseline BMI [kg/m²]		
Mean (SD)	25.2 (4.21)	25.2 (4.21)
Median [Min, Max]	24.8 [13.7, 40.2]	24.8 [13.7, 40.2]
Missing	1 (0.595%)	1 (0.595%)
Baseline eGFR (CKD-2021) [mL/min/1.73 m²]		

	Study	
	CDISCPLOT01 (n=168)	Total (n=168)
Mean (SD)	54.2 (10.6)	54.2 (10.6)
Median [Min, Max]	53.6 [28.4, 87.2]	53.6 [28.4, 87.2]
Missing	2 (1.19%)	2 (1.19%)

Abbreviations: BMI = body mass index, CKD = chronic kidney disease,

eGFR = estimated glomerular filtration rate, max = maximum, min = minimum, SD = Standard Deviation

Note: Missing row appears only when there are missing values

Correlation between covariates

```

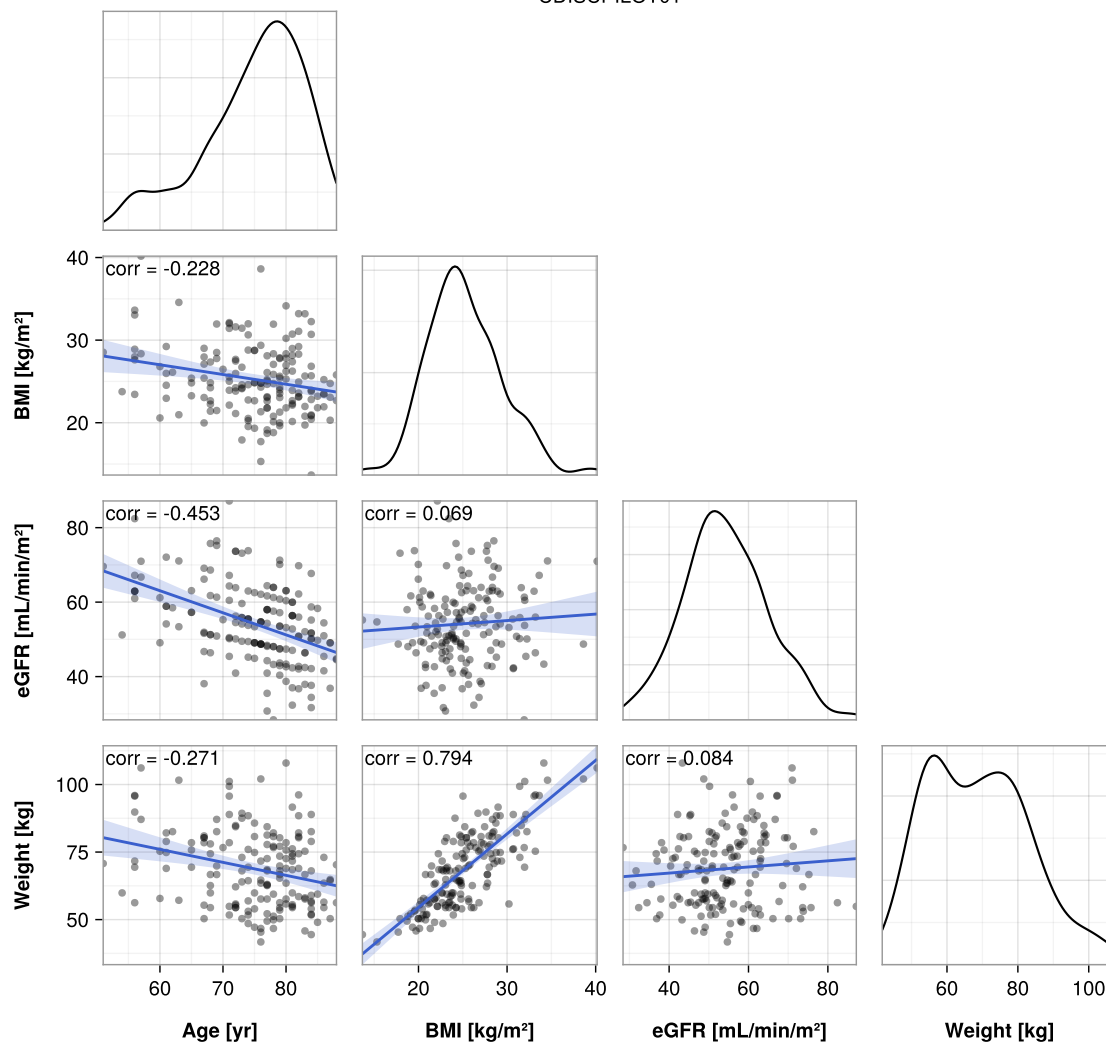
pair_plot = @chain adppk begin
  @select :STUDYIDN :USUBJID :AGE :BMIBL :EGFRBL :WTBL # select only
  covariate data & unique identifiers
  unique # filter to keep only unique
  @select :AGE :BMIBL :EGFRBL :WTBL
  @rename begin # Renaming the columns
    $("Age [yr]") = :AGE
    $(":BMI [kg/m²]") = :BMIBL
    $(":eGFR [mL/min/m²]") = :EGFRBL
    $(":Weight [kg]") = :WTBL
  end
  dropmissing

  # The following function is defined in 20-poppk/julia-functions/pairplot-
  with-regression.
  # Regression Line with CI Band is not available in PairPlots.jl
  pairplot_with_regression(
    color=:royalblue3,
    title="Correlation between Continuous-Valued Covariates",
    subtitle="CDISCPLOT01",
    size=(800, 800)
  )
end

```

Correlation between Continuous-Valued Covariates

CDISCPLOT01



Visualize Covariate Distributions

Continuous variates

```
# Creating a Layer
dist_layer = @chain adppk begin
  @select :STUDYIDN :USUBJID :AGE :BMIBL :EGFRBL :WTBL # select only
  continuous covariate data & unique identifiers
  unique # filter to keep only unique
  @rename begin # Renaming the columns
    $("Age [yr]") = :AGE
    $("BMI [kg/m²]") = :BMIBL
    $("eGFR [mL/min/m²]") = :EGFRBL
```

```

    $(:"Weight [kg]") = :WTBL
  end
  stack(_, [:"Age [yr]", : "BMI [kg/m²]", : "eGFR [mL/min/m²]", : "Weight
[kg]"])
  dropmissing

  data(_) *
  mapping(
    :value,
    layout=:variable,
    color=:variable,
  )

end

```

```

Layer
  transformation: identity
  data: AlgebraOfGraphics.Columns{DataFrames.DataFrameColumns{DataFrame}}
  positional:
    1: value
  named:
    layout: variable
    color: variable

```

Density Plot

```

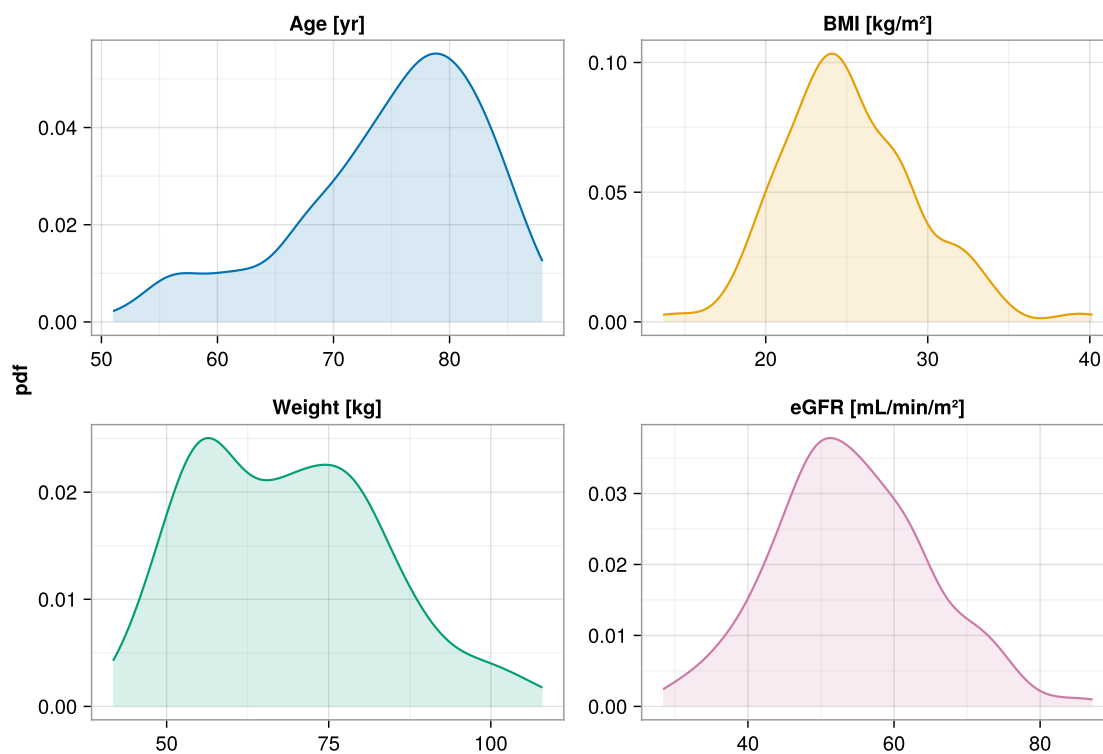
density_plot = dist_layer *
  AlgebraOfGraphics.density(datalimits=extrema)

density_plot = draw(
  density_plot,
  figure=(
    title="Density Plots of Continuous Covariates",
    subtitle="CDISCPIL0T01",
    size=(800, 600)
  ),
  axis=(;
    xlabel="",
  ),
  facet=(
    linkyaxes=:none,
    linkxaxes=:none,
  ),
  legend=(show=false,)
)

```

Density Plots of Continuous Covariates

CDISCPIL0T01

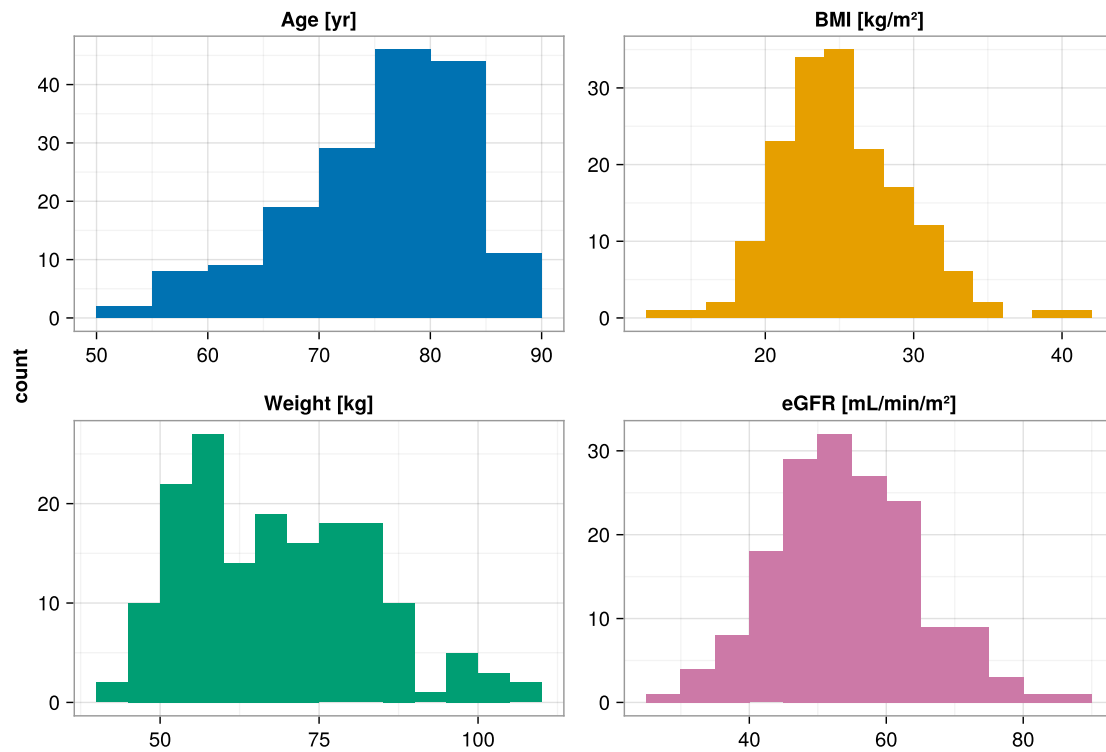


```
hist_plot = dist_layer *
    AlgebraOfGraphics.histogram(bins=15, datalimits=extrema)

hist_plot = draw(
    hist_plot,
    figure=(
        title="Histograms of Continuous Covariates",
        subtitle="CDISCPIL0T01",
        size=(800, 600)
    ),
    axis=(
        xlabel="",
    ),
    facet=(
        linkyaxes=:none,
        linkxaxes=:none,
    ),
    legend=(show=false,)
)
```

Histograms of Continuous Covariates

CDISCPLOT01



Categiocal variates

```
cat_cov = @chain adppk begin
  @select :USUBJID :SEX # Select Categorical covariates
  unique # Getting unique subjects - onw row per subject
  @groupby :SEX
  @combine :n = length(:USUBJID)
  dropmissing

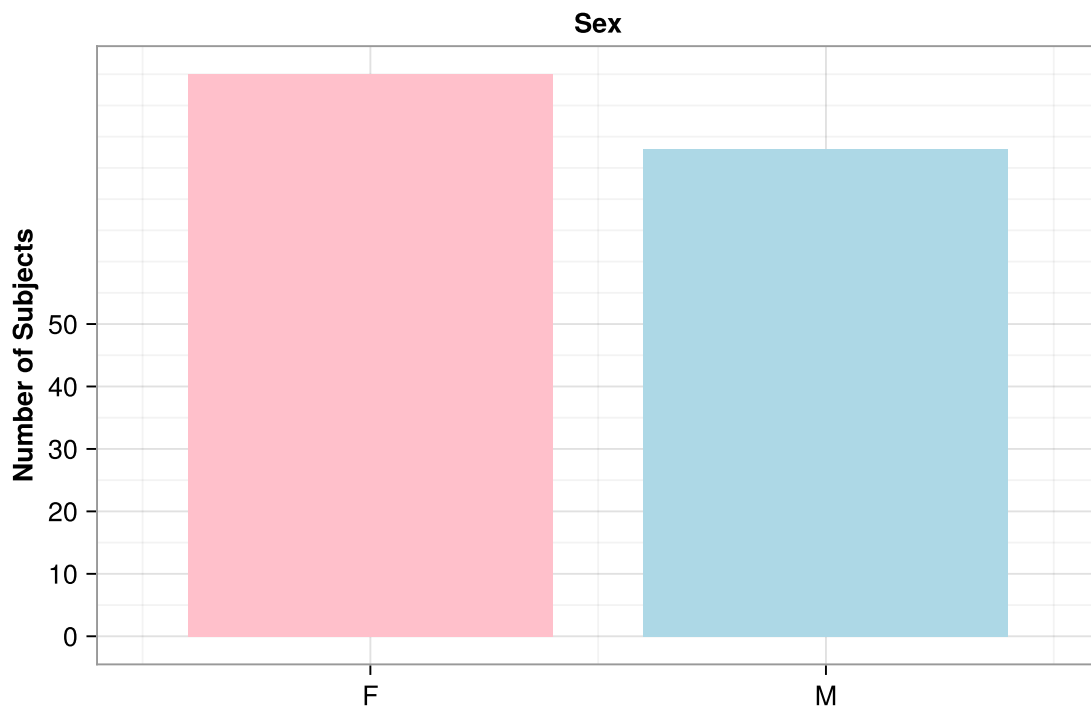
  data(_) *
  mapping(
    :SEX,
    :n,
    color=:SEX
  ) *
  visual(BarPlot,
    color=:lightblue
  )
end
```

```

draw(cat_cov,
  figure=(;
    title="Barplots of Categorical Covariates",
    subtitle="CDISCPIL0T01"
  ),
  axis=(;
    title="Sex",
    yticks=0:10:50,
    ylabel="Number of Subjects",
    xlabel=""
  ),
  scales(;
    Color=(;
      palette=[:pink, :lightblue]
    )
  ),
  legend=(;
    show=false
  )
)

```

Barplots of Categorical Covariates
CDISCPIL0T01



Tabular Summary of assinged treatments

```
adata_subj = unique(adppk, :USUBJID) # Subset to one row per subject (USUBJID)
adata_subj.TX = categorical(adata_subj.COHORTC,
levels=unique(adata_subj.COHORTC))

table_one_mod( # bespoke function table_one_mod - renders Total column in the
last. Use table_one to get it in the front.
  adata_subj,
  [
    :COHORTC => stats_cat => "Actual Treatment, n (%)"
  ],
  groupby=:STUDYID => "Study",
  show_n=true,
  # sort = true
  # footnotes=["Abbreviations: BMI = body mass index, CKD = chronic kidney
disease, eGFR = estimated glomerular filtration rate, max = maximum, min =
minimum, SD = Standard Deviation"]
)
```

	Study	
	CDISCPLOT01 (n=168)	Total (n=168)
Actual Treatment, n (%)		
Xanomeline High Dose	84 (50.0%)	84 (50.0%)
Xanomeline Low Dose	84 (50.0%)	84 (50.0%)

Tabular Summary of BLOQ by Nominal Time relative to previous dose

```
blq_smr = @chain adppk begin
  @transform(:NFRLTC = @. string(round(:NFRLT, digits = 2)))
  @rsubset :EVID == 0
  @groupby :NFRLTC
  @combine begin
    :N = length(:DV)
    :NBLQ = sum(:BLQFN)
    :BLQ [%] = round(sum(:BLQFN) / length(:DV) * 100, digits = 1)
  end
end

total_n = sum(blq_smr.N)
total_nblq = sum(blq_smr.NBLQ)
total_pc_nblq = round(sum(blq_smr.NBLQ) / total_n * 100, digits=1)

# Formatting the TOTAL row to bold and making a DataFrame
```

```

blq_tot_smr = DataFrame([
  :NFRLTC => [styled"{bold:Total}"],
  :N => [styled"{bold:$total_n}"],
  :NBLQ => [styled"{bold:$total_nblq}"],
  Symbol("BLQ [%]") => [styled"{bold:$total_pc_nblq}"]
])

@chain vcat(blq_smr, blq_tot_smr) begin
  @rtransform begin
    :NBLQ = :NBLQ == 0 ? "-" : :NBLQ
    : "BLQ [%]" = : "BLQ [%]" == 0 ? "-" : : "BLQ [%]"
  end

  simple_table(
    [
      :NFRLTC => Annotated("Nominal Sampling Time [hr]", "Relative to
previous dose", label="a"),
      :N => Annotated("N", "Number of Samples", label="b"),
      :NBLQ => Annotated("NBLQ", "Number of BLQ samples", label="c"),
      Symbol("BLQ [%]")
    ],
    halign=:left, :right, :right, :right]
  )
end

```

Nominal Sampling Time [hr] ^a	N ^b	NBLQ ^c	BLQ [%]
0.0	168	168	100
0.08	168	-	-
0.5	168	-	-
1.0	168	-	-
1.5	168	-	-
2.0	168	-	-
4.0	168	-	-
6.0	168	-	-
8.0	168	-	-
12.0	168	-	-
16.0	168	-	-
24.0	168	-	-
36.0	168	168	100
48.0	168	168	100

Nominal Sampling Time [hr] ^a	N ^b	NBLQ ^c	BLQ [%]
Total	2352	504	21.4

^aRelative to previous dose

^bNumber of Samples

^cNumber of BLQ samples

Histograms of samples per subject

```

samples_hist = @chain adppk begin
  # @rsubset :EVID == 0
  @groupby :USUBJID :STUDYID
  @combine :NOBS = length(:DV)
  dropmissing

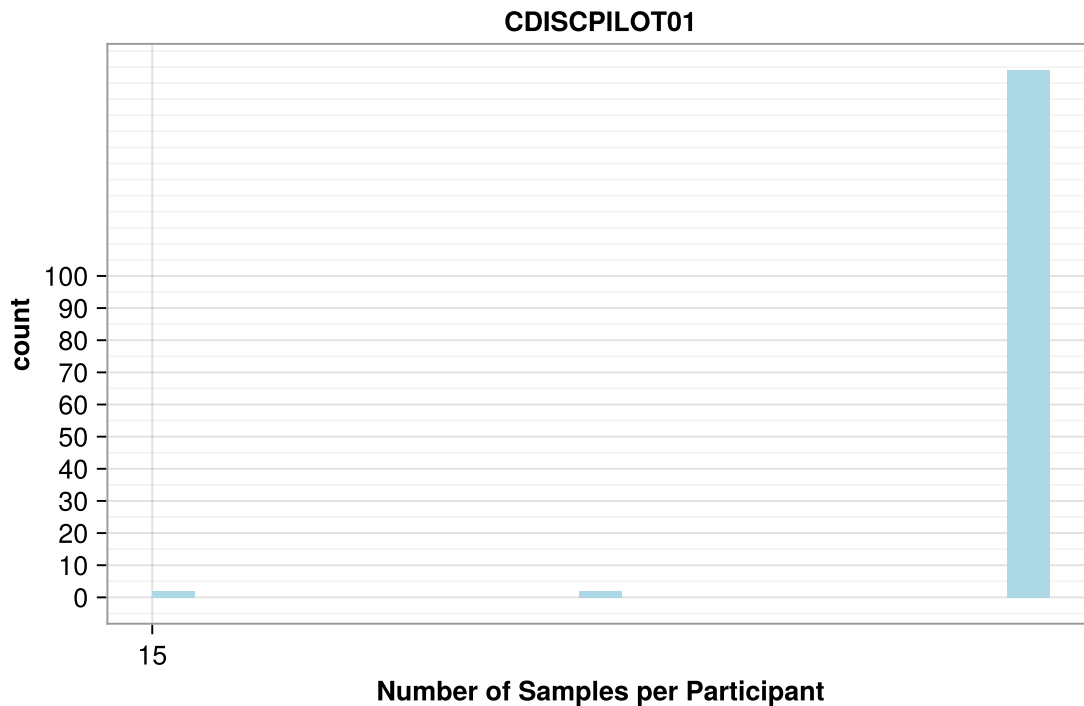
  data(_) *
  mapping(
    :NOBS,
    layout=:STUDYID
  ) *
  AlgebraOfGraphics.histogram(bins=25) *
  visual(color=:lightblue)
end

draw(
  samples_hist,
  figure=(
    title="Histogram of the Number of PK samples, by study",
    subtitle="CDISCIPL0T01"
  ),
  axis=(
    yticks=0:10:100,
    xticks=0:5:100,
    xlabel="Number of Samples per Participant"
  )
)

```

Histogram of the Number of PK samples, by study

CDISCPILLOT01



Histograms of Sampling Times

Relative to previous dose

```
dose_freq_lookup = DataFrame(
    CDISC_VALUE = ["QD", "BID", "TID", "QID"],
    HOURS_BETWEEN_DOSES = [24, 12, 8, 6]
)
```

```
4x2 DataFrame
  Row | CDISC_VALUE  HOURS_BETWEEN_DOSES
     | String      Int64
-----|-----
  1  | QD           24
  2  | BID          12
  3  | TID           8
  4  | QID           6
```

```
adppk = @chain adppk begin
  leftjoin(
```

```

    dose_freq_lookup,
    on = :HOURS_BETWEEN_DOSES,
    matchmissing = :equal
  )
  rename(:CDISC_VALUE => :FREQC)
  # @orderby([:STUDYID, :USUBJID, :EVID])

end
first(adppk,5)

```

```

5x64 DataFrame
 Row | STUDYID      USUBJID      ADY    EVID    NFRLT    HOURS_BETWEEN_DOSES
...  |
...  | String15     String15     Int64  Int64   Float64  Int64?
...  |
-----|-----
  1 | CDISCPIL0T01 01-710-1002      1      1      0.0           24
...  |
  2 | CDISCPIL0T01 01-710-1002      2      1     24.0           24
  3 | CDISCPIL0T01 01-710-1002      3      1     48.0           24
  4 | CDISCPIL0T01 01-717-1004      1      1      0.0           24
  5 | CDISCPIL0T01 01-717-1004      2      1     24.0           24
...  |
                                         58 columns

omitted

```

```

adppk = sort(adppk,[:STUDYID, :USUBJID, :EVID])
first(adppk,5)

```

```

5x64 DataFrame
 Row | STUDYID      USUBJID      ADY    EVID    NFRLT    HOURS_BETWEEN_DOSES
...  |
...  | String15     String15     Int64  Int64   Float64  Int64?
...  |
-----|-----
  1 | CDISCPIL0T01 01-701-1028      0      0      0.0           missing
...  |
  2 | CDISCPIL0T01 01-701-1028      1      0     0.08           missing
  3 | CDISCPIL0T01 01-701-1028      1      0      0.5           missing
  4 | CDISCPIL0T01 01-701-1028      1      0      1.0           missing
  5 | CDISCPIL0T01 01-701-1028      1      0      1.5           missing
...  |
                                         58 columns

omitted

```

```

dist_apr1t = @chain adppk begin
  @rsubset begin
    :EVID == 0
    # :BLQFL == 0
  end
  # dropmissing

  data(_) *
  mapping(
    :APRLT,
    # row=:FREQC => sorter(["QD", "BID", "TID", "QID", missing])
  ) *
  AlgebraOfGraphics.histogram(bins=35, datalimits=extrema) *
  visual(color=:lightblue)

  draw(
    _,
    figure=(
      title="Histogram of Actual Time from First Dose, by Dosing
Frequency",
      subtitle="CDISCP1LOT01",
      size=(800, 600)
    ),
    axis=(;
      # xticks = IntervalsBetween(3)
    ),
    facet=(
      linkyaxes=:none,
      linkxaxes=:none
    )
  )
end

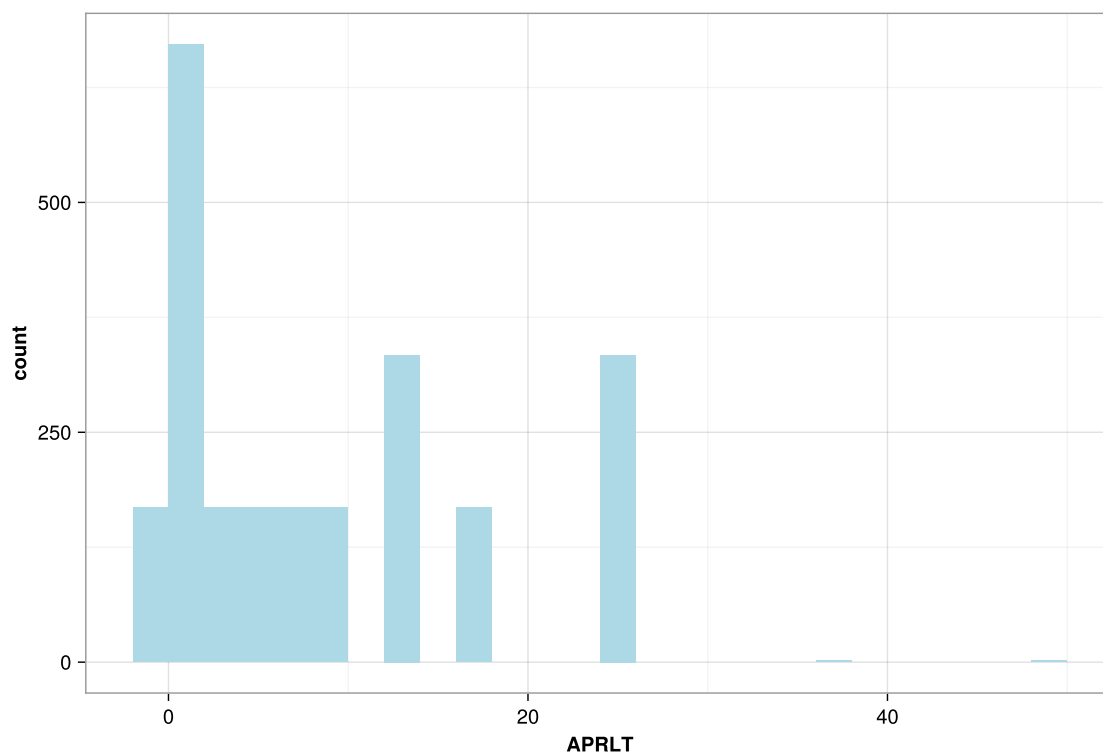
# axes = dist_apr1t.figure.content

# axes[1, 1].xticks = 0:6:100
# axes[2, 1].xticks = 0:2:100
# axes[3, 1].xticks = 0:1:100

dist_apr1t

```

Histogram of Actual Time from First Dose, by Dosing Frequency
CDISCPLOT01



Relative to First Dose

```
dist_nfrlt=@chain adppk begin
  @transform :NFRLTD = :NFRLT / 24
  @rsubset begin
    :EVID == 0
    # :BLQFL == 0
  end

  data(_) *
  mapping(
    :NFRLTD,
    # row=:FREQC => sorter(["Single Dose", "QD", "BID"])
  ) *
  AlgebraOfGraphics.histogram(bins=35, datalimits=extrema,
normalization=:none) *
  visual(color=:lightblue)

  draw(,
    figure=(
      title="Histogram of Nominal Time from First Dose, by Dosing
```

```

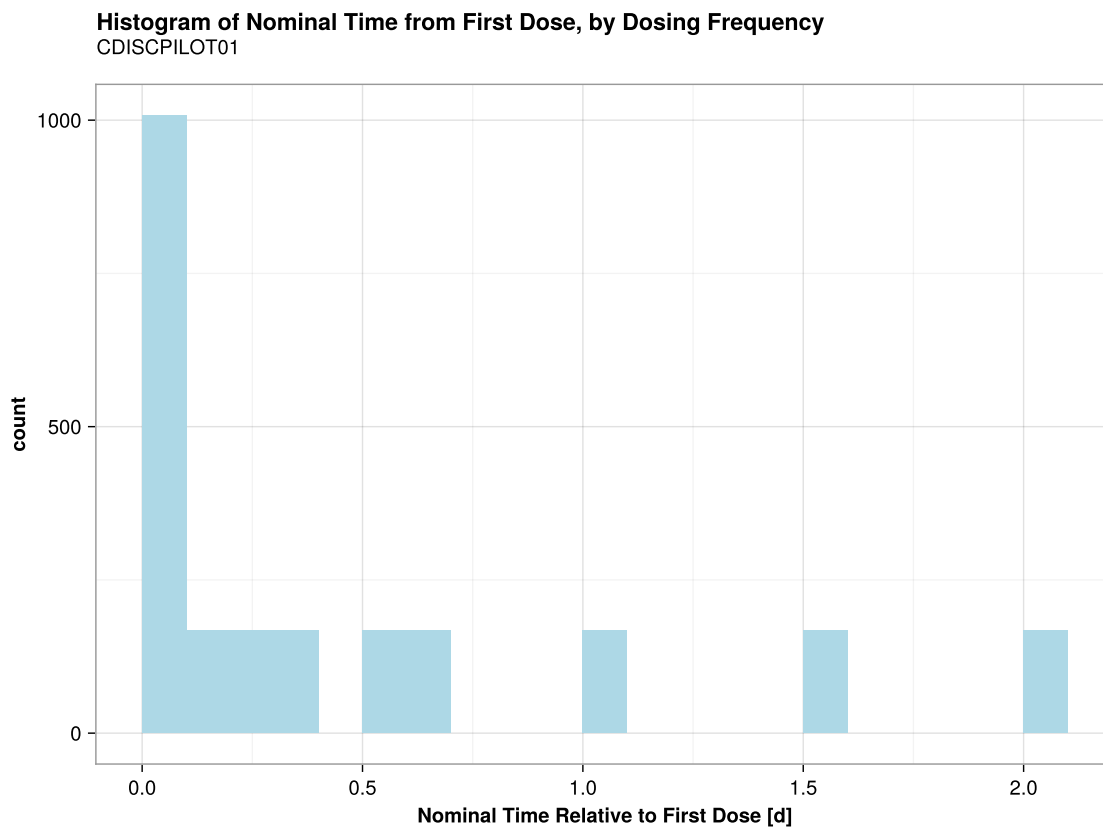
Frequency",
    subtitle="CDISCPIL0T01",
    size=(800, 600)
),
axis=(;
# xticks = 0:2:100,
xlabel = "Nominal Time Relative to First Dose [d]"
),
facet=(
    linkyaxes=:none,
    linkxaxes=:none
)
)
end

# axes = dist_nfrlt.figure.content

# axes[1, 1].xticks = 0:0.5:100
# axes[2, 1].xticks = 0:2:100
# axes[3, 1].xticks = 0:2:100

dist_nfrlt

```



Derive Plotting Data

```
pdata = @chain adppk begin
  @rsubset :STUDYID == "CDISCPIL0T01"

  @transform :TXA = categorical(
    ismissing(:FREQC) ? missing :
    replace(:FREQC, r" \((QD\)" => "QD")
  )

  @rtransform :DNDV = :DV / :DOSEA # dose-normalized DV

  @select :USUBJID :NFRLT :AFRLT :EVID :AMT :DV :DNDV :BLQFL :FREQC :TXA :DOSEA
end
```

```
2850x11 DataFrame
 Row | USUBJID      NFRLT    AFRLT      EVID    AMT      DV      DN
 ... |
   | String15    Float64  Float64?  Int64   Float64?  Float64?  Fl
 ...
```

1	01-701-1028	0.0	-0.5	0	missing	0.0	mi
...							
2	01-701-1028	0.08	0.0833333	0	missing	0.101566	
3	01-701-1028	0.5	0.5	0	missing	0.546902	
4	01-701-1028	1.0	1.0	0	missing	0.925465	
5	01-701-1028	1.5	1.5	0	missing	1.18751	
...							
6	01-701-1028	2.0	2.0	0	missing	1.36889	
7	01-701-1028	4.0	4.0	0	missing	1.68315	
8	01-701-1028	6.0	6.0	0	missing	1.75529	
:	:	:	:	:	:	:	:
2844	01-718-1427	16.0	16.0	0	missing	0.173883	
...							
2845	01-718-1427	24.0	24.0	0	missing	0.015885	
2846	01-718-1427	36.0	36.0	0	missing	missing	mi
2847	01-718-1427	48.0	48.0	0	missing	missing	mi
2848	01-718-1427	0.0	0.0	1	54.0	missing	mi
...							
2849	01-718-1427	24.0	24.0	1	54.0	missing	mi
2850	01-718-1427	48.0	48.0	1	54.0	missing	mi
					5 columns and 2835 rows		
omitted							

```

safe_median(x) = isempty(skipmissing(x)) ? missing : median(skipmissing(x))

pdata_smr = @chain pdata begin
  @rsubset(:EVID .== 0)
  @groupby(:TXA, :NFRLT)
  @combine(
    :DVP50 = safe_median(:DV),
    :DNDVP50 = safe_median(:DNDV),
  )
end

```

14x4 DataFrame				
Row	TXA	NFRLT	DVP50	DNDVP50
	Cat...?	Float64	Float64?	Float64?
1	missing	0.0	0.0	missing
2	missing	0.08	0.100434	0.00185989
3	missing	0.5	0.543987	0.0100738
4	missing	1.0	0.926472	0.0171569
5	missing	1.5	1.19814	0.0221877
6	missing	2.0	1.38567	0.0256606
7	missing	4.0	1.73029	0.0320424
8	missing	6.0	1.81767	0.0336605

9		missing	8.0	1.83799	0.0340368
10		missing	12.0	0.549673	0.0101791
11		missing	16.0	0.162981	0.00301818
12		missing	24.0	0.0142668	0.000264201
13		missing	36.0	missing	missing
14		missing	48.0	missing	missing

Defining plot layers

```
individual_lines =
  data(pdata) *
  mapping(
    :AFRLT,
    :DV,
    group=:USUBJID
  ) *
  visual(
    Lines,
    alpha = 0.5,
    color=:gray75
  )

lloq_line =
  mapping(0) * # LLOQ = 5
  visual(
    HLines,
    linestyle=:dash,
    color=:red,
    linewidth=1,
    label="LLOQ (0 ng/mL)"
  )

rug_nfrlt =
  mapping(unique(pdata_smr.NFRLT)) *
  visual(VLines; color=:red, ymin=0, ymax=0.05, linewidth=1)

median_layer =
  data(pdata_smr) *
  mapping(
    :NFRLT,
    :DVP50
  ) *
  visual(Lines)

dn_median_layer =
  data(pdata_smr) *
  mapping(
```

```

      :NFRLT,
      :DNDVP50
    ) *
    visual(Lines)

median_grouping =
  mapping(
    color=:TXA => sorter(unique(pdata_smr.TXA)),
    linestyle=:TXA => sorter(unique(pdata_smr.TXA))
  )

layout_by_TXA =
  mapping(layout=:TXA => sorter(unique(pdata_smr.TXA)))

```

```

Layer
  transformation: identity
  data: Nothing
  positional:
  named:
    layout: :TXA => AlgebraOfGraphics.Renamer{CategoricalVector{Union{Missing,
String}, UInt32, String, CategoricalValue{String, UInt32}, Missing},
Vector{String}}{Union{Missing, CategoricalValue{String, UInt32}}[missing],
["missing"]}

```

Median C-T Profiles

```

with_theme(my_log_theme) do
  median_ct_plot =
    (
      median_layer * median_grouping
      +
      lloq_line
      +
      rug_nfrlt
    ) |>
    draw(
      figure=(
        title="Median Observed Concentration vs Time, by Dose Level",
        subtitle="CDISCPILOT01",
        size=(800, 600)
      ),
      axis=(ylabel="Plasma Concentration [ng/mL]",
        limits=(nothing, nothing, nothing, nothing)
      ),
      legend=(
        position=:bottom,
        nbanks=2,

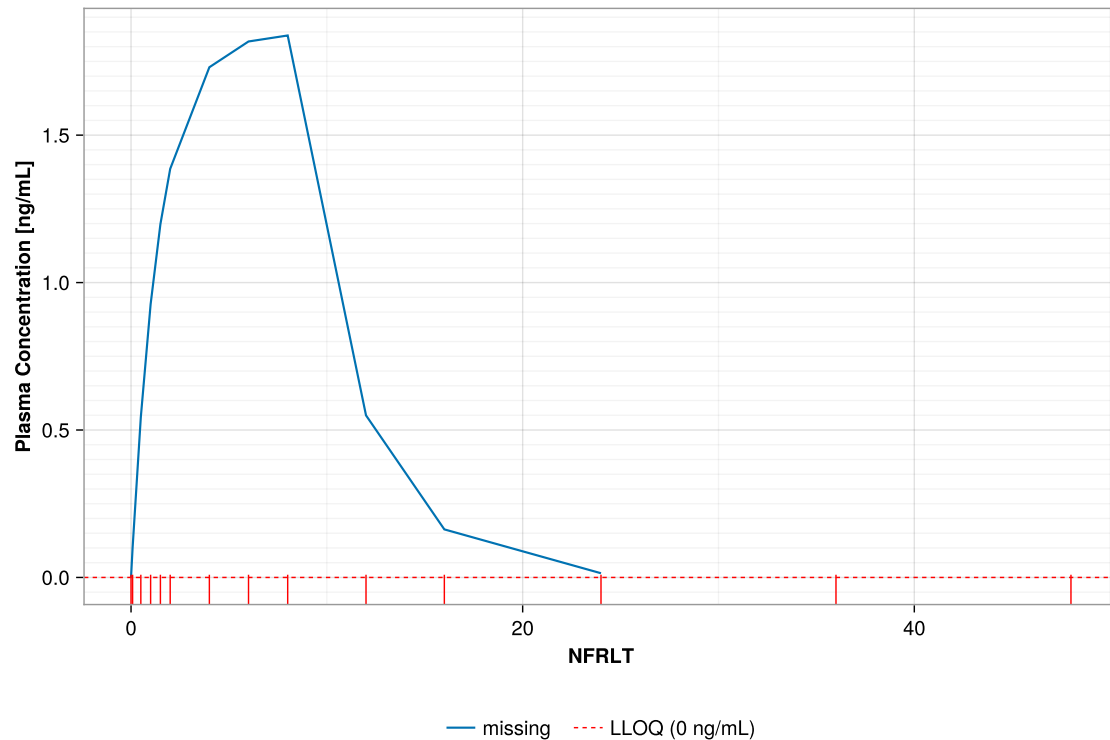
```

```

        framevisible=false,
        titlesize=0
    )
end

```

Median Observed Concentration vs Time, by Dose Level
CDISCPIL0T01



Dose Normalized Median C-T Profiles

```

with_theme(my_log_theme) do
  dn_median_ct_plot =
    (
      dn_median_layer * median_grouping
      +
      rug_nfrlt
    ) |> draw(
      figure=(
        title="Median Dose-Normalized Observed Concentration vs Time,
by Dose Level",
        subtitle="CDISCPIL0T01",

```

```

        size=(800, 600)
    ),
    axis=(
        ylabel="Plasma Concentration [ng/mL]",
        limits=(nothing, nothing, nothing, nothing)
    ),
    legend=(
        position=:bottom,
        nbanks=2,
        framevisible=false,
        titlesize=0
    )
)
end

```

Median Dose-Normalized Observed Concentration vs Time, by Dose Level
CDISCPLOT01

